

Delphes for $HH \rightarrow b\bar{b}\tau\tau$ NSBI: full production tuned and validated against CMS

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Abstract

The full Delphes production is tuned, ntuplised and validated: $3 \times 5\text{M}$ signal events at $\kappa_\lambda = 0, 1, 5$ and 299M $t\bar{t}$ events, 167GB of analysis ntuples. In the semi-leptonic channel — the one the CMS NSBI result uses — Delphes now reproduces CMS to within a few percent in yield ($30\,491$ vs $28\,181$ at $\kappa_\lambda = 0$) and agrees in shape on m_{HH} , m_{bb} , p_T^{HH} , ΔR_{bb} , $\Delta\phi_{HH}$, $p_T^{H1,H2}$, $\cos\theta^*$, both τ p_T and p_T^{miss} . Two things were found in the course of this validation and are worth the PI's attention: the CMS lepton collections were being read *without* ID or isolation, which both broke the semi-leptonic comparison and biased the lepton scale factors (Sec. 2.1); and the b -jet energy *resolution* is untuned by construction, only its scale (Sec. 3.2). One physics residual remains: Delphes makes too many wide-angle τ pairs, visible as an excess at $\Delta R_{\tau\tau} \simeq 2.5\text{--}3.2$ and as high tails on $m_{\tau\tau}$ and m_{vis} (Sec. 3.1).

1 Where we stand

1.1 Production

Everything downstream of Delphes: samples are never re-made, and every correction is a map measured on a private CMS 2024 NanoAODv15 anchor and transferred. The anchor is the PowhegBugFix campaign, the same one the Delphes signal samples were generated from.

sample	events	size	note
signal $\kappa_\lambda = 0, 1, 5$	$3 \times 5.0\text{M}$	7.3 GB	one k1 column per event
$t\bar{t}$	298.7M	160.2 GB	1240/1244 shards (Sec. 3.3)
total	313.7M	167.5 GB	33 parquet files

Table 1: Merged ntuples on /ceph. Produced by a 1394-job HTCondor campaign, 22.4 min/job, 22.0 GB mean / 26.7 GB peak per job.

The κ_λ value is recorded per event. It is encoded only in the input path and the ntuple schema had no field for it, so a merged signal sample would have been unusable for NSBI — the method is entirely about ratios *between* κ_λ hypotheses. It is now recovered from the production plan at merge time and written as a column.

1.2 Tuning

The τ_h energy correction is the one substantive change since the last note. A multiplicative median scale factor cannot reshape a distribution, and the Delphes τ response carries a one-sided tail that CMS does not have (Fig. 1): at gen $p_T = 20\text{--}30\text{ GeV}$ one Delphes τ in ten had more than twice the median response, against 1.08 on CMS. The correction now samples the anchor response *distribution* per object, which reproduces the CMS response by construction and fixes scale and width together. m_{vis} , $\tau_1 p_T$, $\tau_2 p_T$ and p_T^{miss} all overlay CMS after this change.

2 Validation against CMS

2.1 The lepton definition — a defect found and fixed

The CMS τ -candidate pool was being built from the raw NanoAOD **Electron** and **Muon** collections, with *no* ID and no isolation requirement. Those collections are loose enough to contain jet fakes

quantity	status	how
b-tag efficiency (b , c , light)	✓ tuned	stochastic re-tag from <code>Jet.Floror</code> at anchor UParT-AK4 Medium
b-jet energy <i>scale</i>	✓ tuned	inverted to unity against <code>GenJet</code> (self-anchored), 0.965–0.989
b-jet energy <i>resolution</i>	✗ tuned	not corrected at all — Sec. 3.2
τ_h ID efficiency	✓ tuned	anchor <code>GenVisTau</code> → <code>DeepTau</code> v2.5 Medium; includes HPS reco inefficiency
τ_h energy response	✓ tuned	<i>distribution</i> resampled from anchor quantiles, not a median ratio
τ_h visible mass	✓ tuned	sampled per jet from anchor per- p_T quantiles
p_T^{miss} resolution	✓ tuned	Gaussian smearing binned in jet H_T
lepton reco efficiency	✓ tuned	per- p_T SFs, now 0.96–1.12 (e), 1.03–1.07 (μ)
di- τ mass estimator	✓ identical	same covariance-free FastMTT on both sides

Table 2: Object status. Every map is derived on the CMS anchor and applied downstream; none needs a card change.

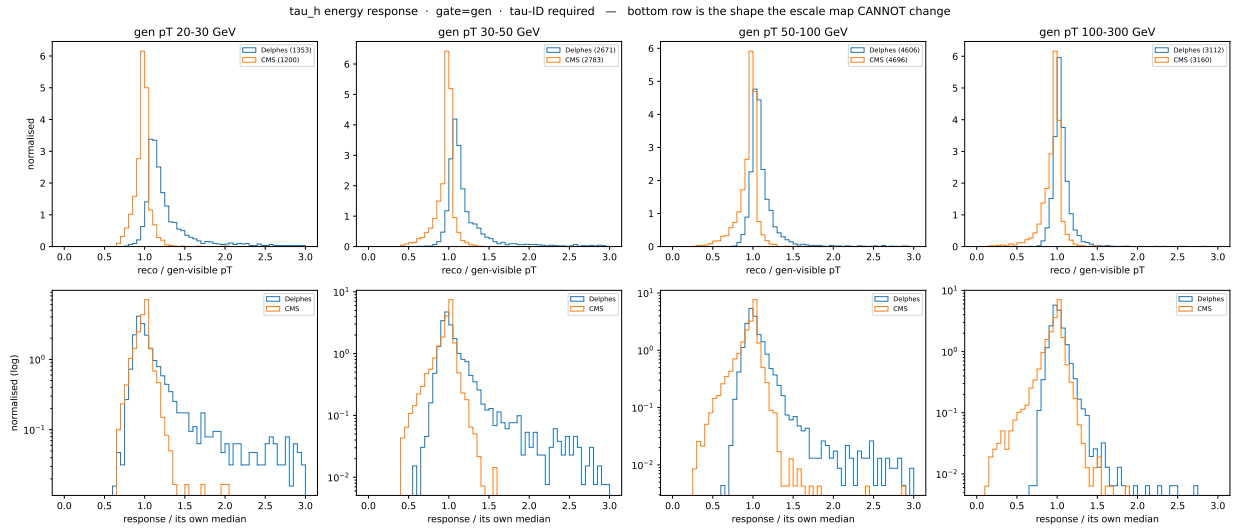


Figure 1: τ_h energy response, Delphes vs the CMS anchor. The Delphes tail above the median is what a multiplicative correction cannot remove and what motivated resampling the distribution instead.

that no analysis would use, whereas a Delphes lepton has already passed the card’s efficiency parameterisation, which is effectively an ID. The two pools therefore did not mean the same thing, and the consequences were not cosmetic:

- CMS acquired candidates Delphes cannot produce — 49 950 selected events against Delphes’ 30 491 at $\kappa_\lambda = 0$, a $1.6\times$ excess read at the time as a Delphes acceptance deficit.
- A fake lepton paired with a real τ is back-to-back, so the CMS $\Delta R_{\tau\tau}$ and ΔR_{bb} distributions grew a spurious second peak at $\Delta R \simeq \pi$, plus $m_{\tau\tau}$ and m_{vis} tails and a flattened $\cos\theta^*$.
- The same collections feed `lepton_efficiency`, which sets the `electron_sf` / `muon_sf` anchor targets. The scale factors were therefore tuned toward loose-reco efficiency rather than toward analysis-quality leptons.

The cut is now applied in the NanoAOD *reader*, so the tuning anchor and the validation overlay cannot use different lepton definitions again, and a configured-but-missing ID branch raises rather than silently reverting to the loose collection. Working points are `Electron_mvIso_WP80`, `Muon_tightId` and `Muon_pfRelIso04_all < 0.15`; **these should be checked against the CMS $b\bar{b}\tau\tau$ baseline before anything is quoted**. After the fix the lepton fractions match (30% Delphes vs 33% CMS) and the scale factors sit near unity, which is itself the evidence that the two definitions now agree — against the loose collection they had to make up a $\sim 20\%$ gap.

2.2 Result

✓ agree	m_{HH} (the κ_λ observable, at $\kappa_\lambda = 0, 1, 5$); m_{bb} peak; p_T^{HH} ; ΔR_{bb} ; $\Delta\phi_{HH}$; p_T^{H1}, p_T^{H2} ; $\cos\theta^*$; $\tau_1 p_T, \tau_2 p_T$; p_T^{miss}
✗ disagree	$\Delta R_{\tau\tau}$ (Delphes excess at 2.5–3.2), and the $m_{\tau\tau}$ / m_{vis} high tails that follow from it; m_{bb} width

Table 3: NSBI inputs, semi-leptonic channel, 200k events per κ_λ point against the `PowhegBugFix` anchor including its `ext1` extension. Selected yields Delphes/CMS: 30 491/28 181 ($\kappa_\lambda = 0$), 32 783/31 442 ($\kappa_\lambda = 1$), 24 218/20 443 ($\kappa_\lambda = 5$).

3 Open items

3.1 $\Delta R_{\tau\tau}$: too many wide-angle τ pairs

The one physics residual. Delphes has an excess at $\Delta R_{\tau\tau} \simeq 2.5\text{--}3.2$, and the $m_{\tau\tau}$ and m_{vis} high tails are probably the same events counted twice — but that is an inference, not a demonstration, so the argument is worth stating. For legs whose masses are negligible against their energies, $m_{\text{vis}}^2 = 2 p_{T1} p_{T2} (\cosh\Delta\eta - \cos\Delta\phi)$ exactly, and both angular terms grow monotonically as the pair separates; $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ inherits it. A wide pair is therefore a heavy pair — though m_{vis} is monotonic in $\Delta\eta$ and $\Delta\phi$ separately and is *not* a function of ΔR alone, so “wide-angle” rather than “back-to-back” is what ΔR measures. The converse does not follow by itself: a heavy tail could equally come from harder τ , since m_{vis} also grows with $p_{T1} p_{T2}$. What excludes that here is that $\tau_1 p_T$ and $\tau_2 p_T$ both overlay CMS, leaving the angular term as the only place the excess mass can come from. The direct test — cutting $\Delta R_{\tau\tau} < 2.5$ and asking whether the m_{vis} and $m_{\tau\tau}$ tails go with it — has not been run; if the tails survive, these are two problems and not one. This matters for the feature set: dropping $\Delta R_{\tau\tau}$ from the NSBI inputs would remove the most visible symptom while leaving the identical artefact in $m_{\tau\tau}$ and m_{vis} , which we certainly keep. The cause has to be found rather than worked around.

The leading hypothesis is τ_h mistag: a fake τ from the opposite hemisphere pairs with a real one at $\Delta R \simeq \pi$. This is directly testable — the overlay can split each side into gen-matched and

fake τ pairs and report the fake fraction — and has not yet been run at full statistics. It is also actionable if confirmed: the `tau_fake_response` map has its top three p_T bins below the 200-entry threshold and is borrowing them from neighbours, so the high- p_T fake response is extrapolated rather than measured.

3.2 m_{bb} width: the b -jet resolution is untuned

The b -jet correction inverts the response to unity against Delphes’ *own* GenJet. That fixes the scale, does not touch the width, and never references CMS at all. The τ side, by contrast, now has its full response distribution resampled from the CMS anchor. m_{bb} is therefore the feature receiving the weaker of the two treatments, and its width does not match CMS.

The fix is the direct analogue of what worked for τ : measure the CMS b -jet response as a distribution and resample it. The machinery exists. The cost is the decision — it requires re-deriving the maps and re-running the full campaign (~ 10 h) — and it should be validated on a subset before that is spent. Note also that the card runs without pileup, so part of the CMS b -jet resolution has no counterpart in Delphes to correct.

3.3 Four lost $t\bar{t}$ shards

Four of 1244 $t\bar{t}$ shards (0.3%) could not be produced: the dCache redirector consistently routes those files to a pool node whose certificate does not cover the address it hands out. Bounded retries were added and confirm the failure is deterministic, not transient, so this is a site issue to report rather than a pipeline defect. Shards group input files by size with no relation to physics, so the loss costs MC statistics and introduces no bias; the shortfall is recorded in the merged manifest rather than hidden.

4 Decisions and next steps

1. **Confirm the lepton working points** against the CMS $b\bar{b}\tau\tau$ baseline. Everything downstream follows from them, and the current values are conventional choices rather than the analysis’s.
2. **Run the gen-matched/fake split at full statistics** to test whether the $\Delta R_{\tau\tau}$ excess is τ mistag. Cheap, and it decides whether Sec. 3.1 is a fixable map problem or something deeper.
3. **Decide whether to tune the b -jet resolution** (Sec. 3.2). This is the only item costing a re-production, so it should be validated on a subset first.
4. **Then train.** The ntuples are ready and κ_λ -labelled; the feature set should be settled by measured κ_λ separation and by training with and without candidate features, not by inspecting one-dimensional marginals.

5 Known limitations

Accepted by design. FastMTT is covariance-free but is run identically on both sides, so it cancels in the comparison and bounds the absolute $m_{\tau\tau}$ resolution rather than the agreement. The card runs without pileup, so jet energy resolution is untuned and pileup jets cannot exist; neither is fixable downstream. p_T^{miss} smearing is now binned in jet H_T but remains isotropic where the real resolution is anisotropic.

A scope bound, not a defect. `tau_mistag` is measured on the b -rich signal anchor and fake rates are strongly jet-flavour dependent, so it is not valid for fake-dominated backgrounds (QCD, W +jets). The present signal study is unaffected; a background-heavy analysis would need those measured separately. There is likewise no CMS anchor for DY, so DY cannot currently be tuned.

Untested — risks, not accepted limitations. The sampled τ_h mass is drawn independently of the τ energy and of the partner leg, whereas the true visible mass correlates with decay mode; size unknown. Every map is flat-extrapolated above 300 GeV. That is probably benign here — only $\sim 2.8\%$ of b-jets exceed it, the card’s b-tag formula tends to a constant so flat is the right asymptote, and κ_λ discrimination lives near threshold ($m_{HH} \simeq 250\text{--}400$ GeV, where the triangle’s s -channel propagator makes the interference most κ_λ -sensitive) rather than in the box-dominated tail. A boosted or high- m_{HH} category would change that, by making the extrapolated region the signal region.